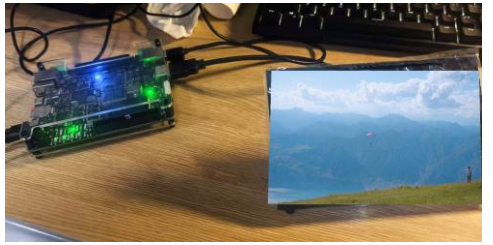
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One Frame, Two Truths
A Multi-Source Video Node on Zynq-7020

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Platform: RK-ZYNQ7020-F · xc7z020clg484-2 · Vivado/Vitis 2025.2.1



On-board test · system output frame

INTRODUCTION

Remote video nodes usually have one Ethernet cable and no PC. The operator needs to know whether the frame arrived, whether processing took effect, and how old the picture is. This node performs the complete path on one Zynq-7020: a 512×300 RGB565 stream enters through GbE, whole frames are committed after three conditions, both panes read back the same frame, and the seam selects the original or processed pixel.

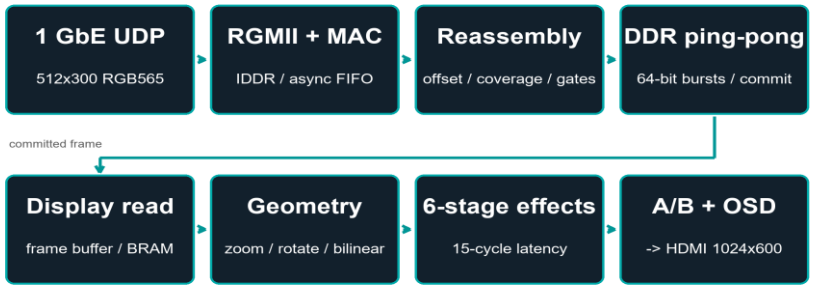
Typical application case

Confidence monitoring for broadcast, security and industrial vision: one cable carries the picture, processing result and link status.

Key technical issues addressed

1. GbE UDP reassembly: offset-based reassembly; commit only when bytes, row coverage and bad-packet count pass.
2. Same-frame comparison: both panes share geometry and a fixed 15-cycle effect chain; the seam selects per pixel.
3. Status monitoring: aborts, missing rows, frame gaps and latency are readable through serial, telemetry and GPIO.
4. Storage: DDR ping-pong frame buffer + on-board test card.

SYSTEM ARCHITECTURE



1. GbE UDP reception & reassembly: RGMII → MAC → UDP filter → whole-frame commit;
2. DDR ping-pong buffer: PL as AXI HP0 master, double-buffered;
3. Geometry engine: 0.25–2.00× zoom (8 steps) + 0–359° rotation + bilinear;
4. 15-cycle effect chain: gray/invert/blur/sharpen/Sobel/binary/erode-dilate, constant latency;
5. Same-frame comparison: left original, right processed, per-pixel seam;
6. OSD + status readback: serial, telemetry and GPIO read the same counters.

CREATIVE DESIGN

1. Same-frame comparison

Both panes share geometry, row labels and the fixed 15-cycle chain — the seam compares one frame, not two screenshots.

2. Commit conditions + status monitoring

Bytes, row coverage and bad-packet count must pass together; aborts, missing rows and frame gaps explain failures at 1/100 drops.

RESULT

RGMII → HDMI entirely in PL; PS only control. 1024×600 @ 59.5 Hz.

Timing: WNS 1.368 / WHS 0.048 ns, zero failures.

8 s stream: 241 frames @ 30.01 fps; selftest 22.2/0/18.9 PASS.

UART 105 commands PASS; QSPI boot validated.

Next: DDR tile cache / 1080p link / RGMII eye diagram / remote update.

Test tools: UART battery (105) · link selftest · telemetry listen | Performance: 30.01 fps | Asset: royalty-free scenery video

AMD